Chengzhi Lu

CONTACT INFORMATION

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NTU Singapore

EDUCATION

University of Chinese Academy of Sciences, China, (Sep. 2019 – Jan. 2025)

Ph.D., Computer Science
Advisor: Prof. Chengzhong Xu, Dean of FST, University of Macau

Zhejiang University, China, (Sep. 2016 - Sep. 2018)

M.S., Software Engineering

Wuhan University, China, (Sep. 2012 - Jun. 2016)

B.S., Computer Science

CONFERENCE PUBLICATIONS

[**SoCC'26, Vision**] JooYoung Park, Leonid Kondrashov, Boxi Zhou, Hongrui Liu, Zonghao Liu, **Chengzhi Lu**, Riccardo Mancini, Esha Choukse, Haris Javaid, German Sviridov, Tao Peng, Chen Zhao, Marios Kogias, Luo Mai, Anastasia Avdeeva, Aleksei Gusev, Dmitrii Ustiugov. Rethinking AI Cloud Infrastructure for Agentic Serving Systems with the Aries Experimentation Framework.

[**SoCC'26**] Ruiqi Lai, **Chengzhi Lu**, Hongrui Liu, Siyu Cao, Li Leqi, Zonghao Liu, Siyang Shao, Yixin Zhang, Luo Mai, Dmitrii Ustiugov. TokenScale: Timely and Accurate Autoscaling for Disaggregated LLM Serving with Token Velocity.

[**ICPP'26**] Shijie Peng, Yanying Lin, **Chengzhi Lu**, Shuaipeng Wu, Chengzhong Xu, Kejiang Ye. ReliefServe: Relieving GPU Pressure in Multi-Model Serving via Selective CPU Escape.

[**EuroSys'26**] Wenyan Chen, **Chengzhi Lu**, Huanle Xu, Kejiang Ye, Chengzhong Xu. High Throughput and Low Latency LLM Serving via Adaptive KV Caching.

[**EuroSys'26**] Yanying Lin, Shijie Peng, **Chengzhi Lu**, ChengZhong Xu, and Kejiang Ye. FlexPipe: Adapting Dynamic LLM Serving Through Inflight Pipeline Refactoring in Fragmented Serverless Clusters.

[**EuroSys'25**] Wenyan Chen, **Chengzhi Lu**, Huanle Xu, Kejiang Ye, and Cheng-Zhong Xu. Multiplexing Dynamic Deep Learning Workloads with SLO-awareness in GPU Clusters.

[**SC'24**] **Chengzhi Lu**, Huanle Xu, Yudan Li, Wenyan Chen, Kejiang Ye, and Chengzhong Xu. SMIless: Serving DAG-based Inference with Dynamic Invocations under Serverless Computing.

[**ICWS'24**] Yanying Lin, Shijie Peng, Shuaipeng Wu, Yanbo Li, **Chengzhi Lu**, Chengzhong Xu, Kejiang Ye. Planck: Optimizing LLM Inference Performance in Pipeline Parallelism with Fine-Grained SLO Constraint.

[**EuroSys'23**] **Chengzhi Lu***, Huanle Xu*, Kejiang Ye, Guoyao Xu, Liping Zhang, Guodong Yang, and Chengzhong Xu. Understanding and Optimizing Workloads for Unified Resource Management in Large Cloud Platforms.

[**SoCC'21**] Shutian Luo*, Huanle Xu*, **Chengzhi Lu**, Kejiang Ye, Guoyao Xu, Liping Zhang, Yu Ding, Jian He, Chengzhong Xu. Characterizing Microservice Dependency and Performance: Alibaba Trace Analysis (**Best Paper Award**).

[**CLUSTER'21**] Wenyan Chen, **Chengzhi Lu**, Kejiang Ye, Yang Wang, Chengzhong Xu. RPTCN: Resource Prediction for High-dynamic Workloads in Clouds based on Deep Learning.

[**ICPADS'19**] **Chengzhi Lu**, Kejiang Ye, Wenyan Chen, Chengzhong Xu. ADGS: Anomaly Detection and Localization based on Graph Similarity in Container-based Clouds.

[**ICPADS'18**] Kejiang Ye, Yanmin Kou, **Chengzhi Lu**, Yang Wang, Chengzhong Xu. Modeling Application Performance in Docker Containers Using Machine Learning Techniques.

[**Big Data'17**] **Chengzhi Lu**, Kejiang Ye, Guoyao Xu, Cheng-Zhong Xu, Tongxin Bai. Imbalance in the cloud: an analysis on Alibaba cluster trace.

Journal Publications

[**TSC,2025**] Yanying Lin, Shijie Peng, Shuaipeng Wu, Yanbo Li, **Chengzhi Lu**, Kejiang Ye, Chengzhong Xu. Serving LLM in Distributed GPU Cluster With Fine-Grain Pipeline Constraints.

[**TPDS,2022**] Shutian Luo*, Huanle Xu*, **Chengzhi Lu**, Kejiang Ye, Guoyao Xu, Liping Zhang, Jian He, Chengzhong Xu. An in-depth study of microservice call graph and runtime performance.

[**JCST,2020**] Wen-Yan Chen, Ke-Jiang Ye, **Chengzhi-Lu**, Dong-Dai Zhou, Cheng-Zhong Xu Interference analysis of co-located container workloads: a perspective from hardware performance counters.

Preprints

[**arXiv,2026**] Wenyan Chen, **Chengzhi Lu**, Yanying Lin, Dmitrii Ustiugov. FASER: Fine-Grained Phase Management for Speculative Decoding in Dynamic LLM Serving.

Recent Research Projects

System Optimization for Supporting Efficient AI Applications

- Improve performance of LLM inference service by co-optimizing hardware resources and application configurations.

Cloud Computing Resource Management, Performance Optimization

- Design scheduling algorithms with an optimistic fault tolerance mechanism to improve throughput and cluster resource utilization.

Resource Allocation Algorithms for Large-Scale Clusters with Performance Guarantees

- Design ML-based resource configuration strategy for the large-scale co-located cluster.

Academic Experience

NTU Singapore
Postdoc — Sep. 2025 - Now

- Resource management for LLM serving.

University of Macau
Postdoc — Jan. 2025 - Aug. 2025

- Resource allocation for inference serving in GPU cluster.
- Request scheduling for LLM applications.

University of Macau
Research Assitant — Feb. 2022 - Dec. 2024

- Resource allocation algorithms for large-scale clusters with performance guarantees.
- Resource assignment for serverless applications
- Request scheduling for LLM applications.

Shenzhen Institute of Advanced Technology, Chinese Academy of Sciences
Visting Student — Jul. 2017 - Sep. 2019

- Distributed system resources management, including microservices and DL distributed training.
- Resource allocation algorithms for large-scale clusters with performance guarantees.

- Anomaly detection algorithm for large-scale clusters.
- Server cluster administrator, manages more than 140 physical servers, and provides experimental environment and services including Docker, GPU, Spark, Kubernetes, etc.

Work Experience

Alibaba Group, Zhejiang, China
Academic Intern — Jun. 2020 - Feb. 2023

- Participate in the functional development of Alibaba's infrastructure platform.

Honors and Awards

Dean's award for academic performance, SIAT, Chinese Academy of Sciences — Jan. 2021, 2024
Outstanding student, University of Chinese Academy of Sciences, — Jan. 2022
Outstanding student, SIAT, Chinese Academy of Sciences — Feb. 2019
Outstanding graduate, Zhejiang University — Jul. 2018

Services

Teaching Assistant of Computer Network, SIAT, Chinese Academy of Sciences — 2021

Skills

Python, GO, Git, Latex, Java
Kubernetes, Microservice, Serverless, Docker